

Supporting Information for "A seismologically-consistent expression for the total area and volume of earthquake-triggered landsliding."

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Additional Supporting Information (Files uploaded separately)

1. Captions for large Table S1 (if larger than 1 page, upload as separate excel file)

Introduction

The supporting information contains 2 data tables (one is uploaded as an additional excel file) summarizing the source, value and uncertainties of parameters used in the main text.

Additionally it contains 10 figures that support the conclusions of the main text.

Figure SF 1 shows the 40 earthquakes of our database on a world map.

Figure SF 2 presents the imagery used to map and produce a new earthquake-induced landslide inventory for the 1991 Limon earthquake (Costa Rica).

Figure SF 3 supports one of our model assumptions in showing that horizontal peak ground acceleration and landslide volume density seem linearly correlated.

Figure SF 4 shows the model sensitivity to different parameters assessed through Monte Carlo simulations.

Figure SF 5 complements the main text Figure 3 and shows the slope histograms of all the remaining earthquake areas.

Figure SF 6 shows the final model residuals against depth, landscape steepness, fault type and seismic moment.

Figure SF 7 shows the final model residuals against moment, obtained when we choose different frequencies (0.25 - 1 - 6Hz) for our seismic source scaling constants.

Figure SF 8 shows the model accuracy changes when we consider the hypocenter as the source depth instead of the mean asperity depth.

Figure SF 9 shows the model accuracy changes when we remove the assumption that normal fault earthquakes produce less ground shaking.

Figure SF 10 shows the landslide area-frequency distribution for our 11 exhaustive inventories.

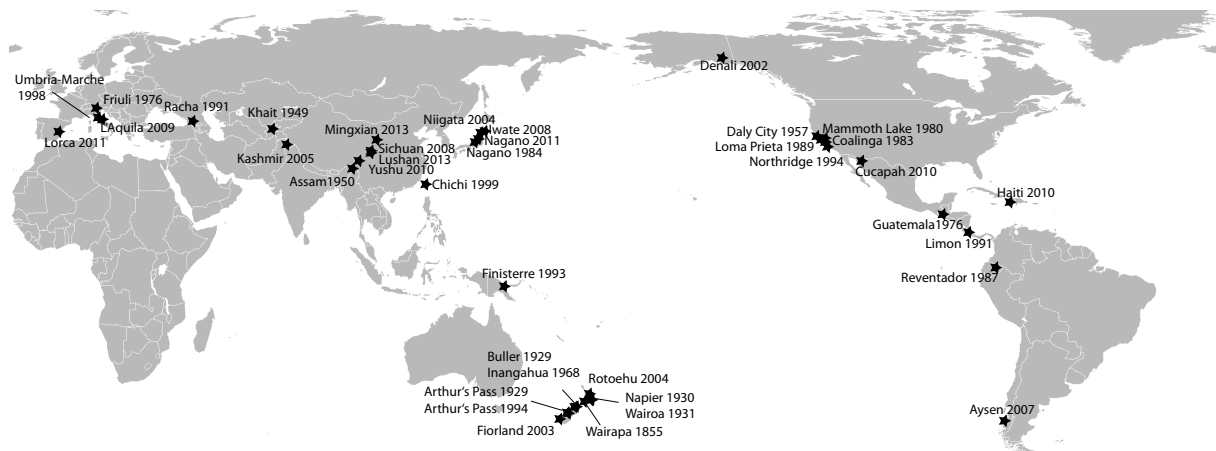


Figure S1. World map with the locations of the 40 shallow earthquakes considered in this study.

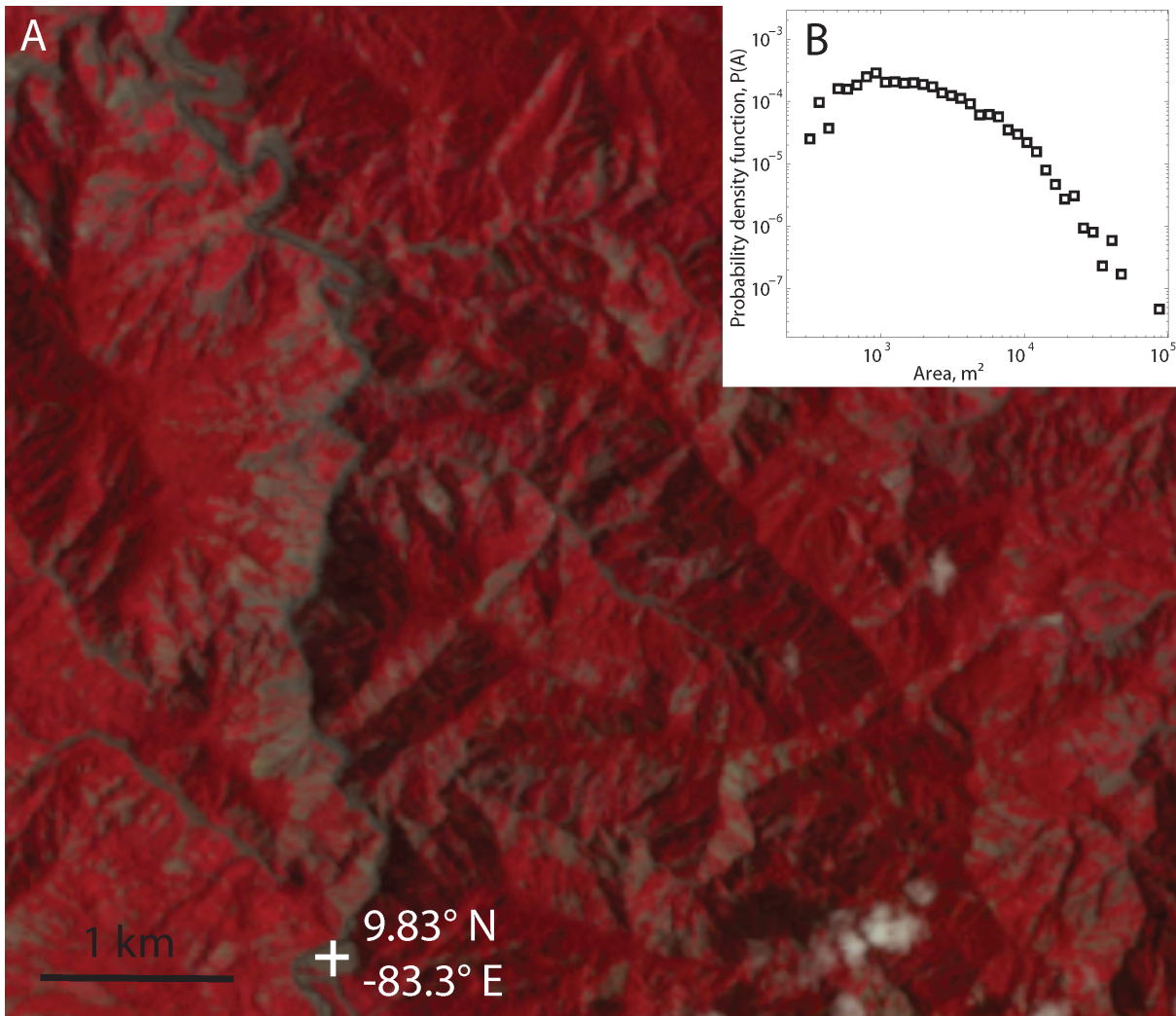


Figure S2. Landsat image (February 1992) of the epicentral area of the 1991 Limon (Costa Rica) earthquake. Red represents vegetation whereas gray and white are bare rock or sediment and clouds, respectively. Visible landslides were mapped down to about 1000m^2 . The area-frequency distribution of 1643 large ($> 1000\text{m}^2$) mapped landslides is shown in the inset. The roll-over at $1000 - 3000\text{m}^2$ is likely due to resolution censoring. The total number of landslide is likely more than 3000.

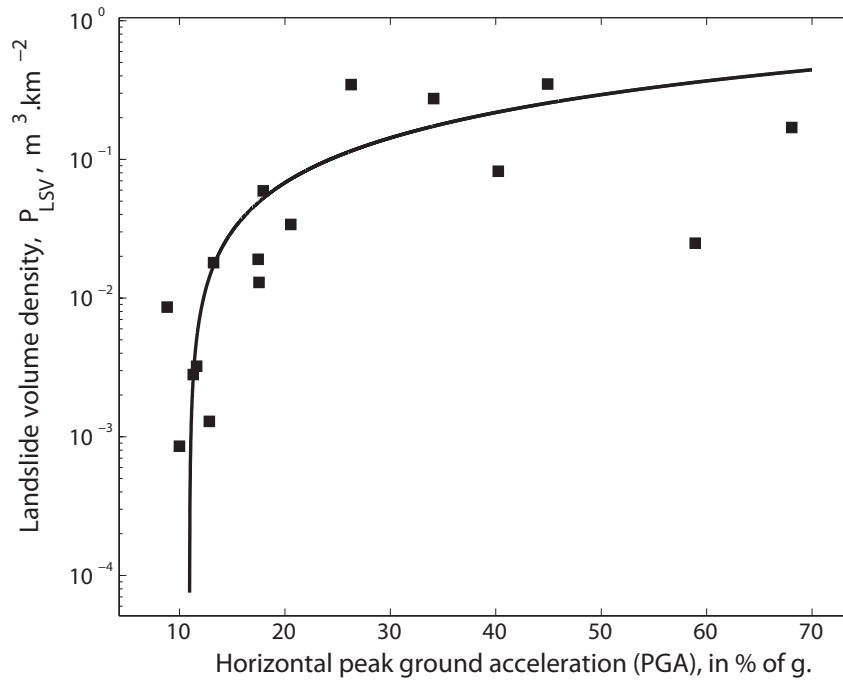


Figure S3. Landslide volume density as a function of the horizontal peak ground acceleration (normalized by $1g$) in the ChiChi epicentral area. Available PGA records from seismic stations in Taiwan and landslide volume density were averaged within circular bins with increasing distance away from the earthquake epicenter. For reference the black curve represents the linear function $Y = 0.75(PGA - 0.11)$.

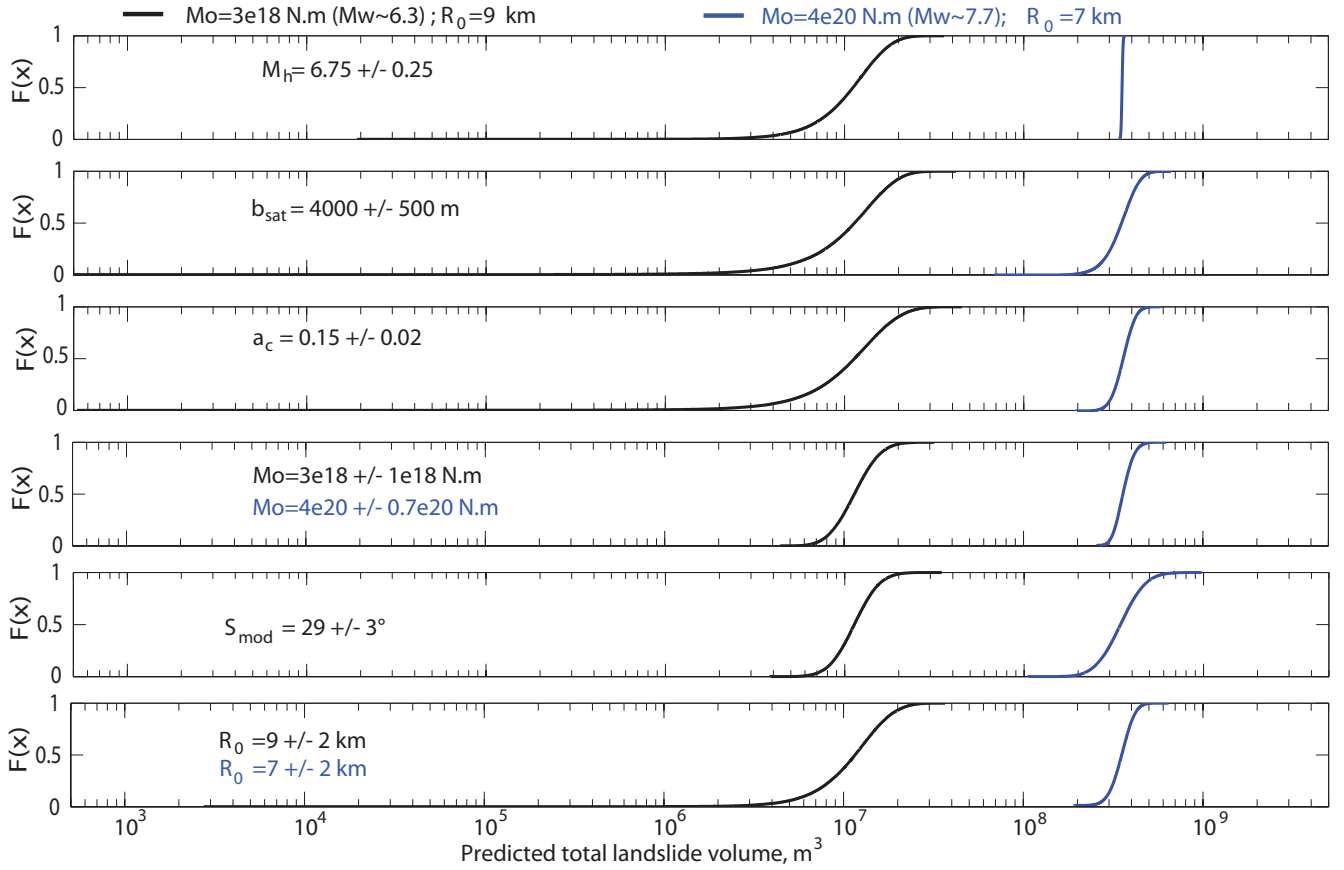


Figure S4. Cumulative distribution of total landslide volume predictions for one relatively small and deep earthquake (in black) and a larger shallower one (in blue). In each panel a single parameter is varied while the others are kept constant at their mean value. Considered are the essential model parameters M_h , the hinge magnitude, b_{sat} , the saturation acceleration and a_c , the critical acceleration. Also considered are the key input data, the seismic moment Mo and the mean asperity depth, R_0 as well as the modal slope S_{mod} used to constrain the landscape sensitivity. In this exploration, the small, deep earthquake is in the very non-linear part of the model (i.e. before M_h) and is therefore much more sensitive to variations of the input parameters or of the model constant than the larger earthquake. Note that the model uncertainty ranges presented in Fig 4-6, are based on the same assumed variations of M_h , b_{sat} and a_c , and on input uncertainties reported in Table S1.

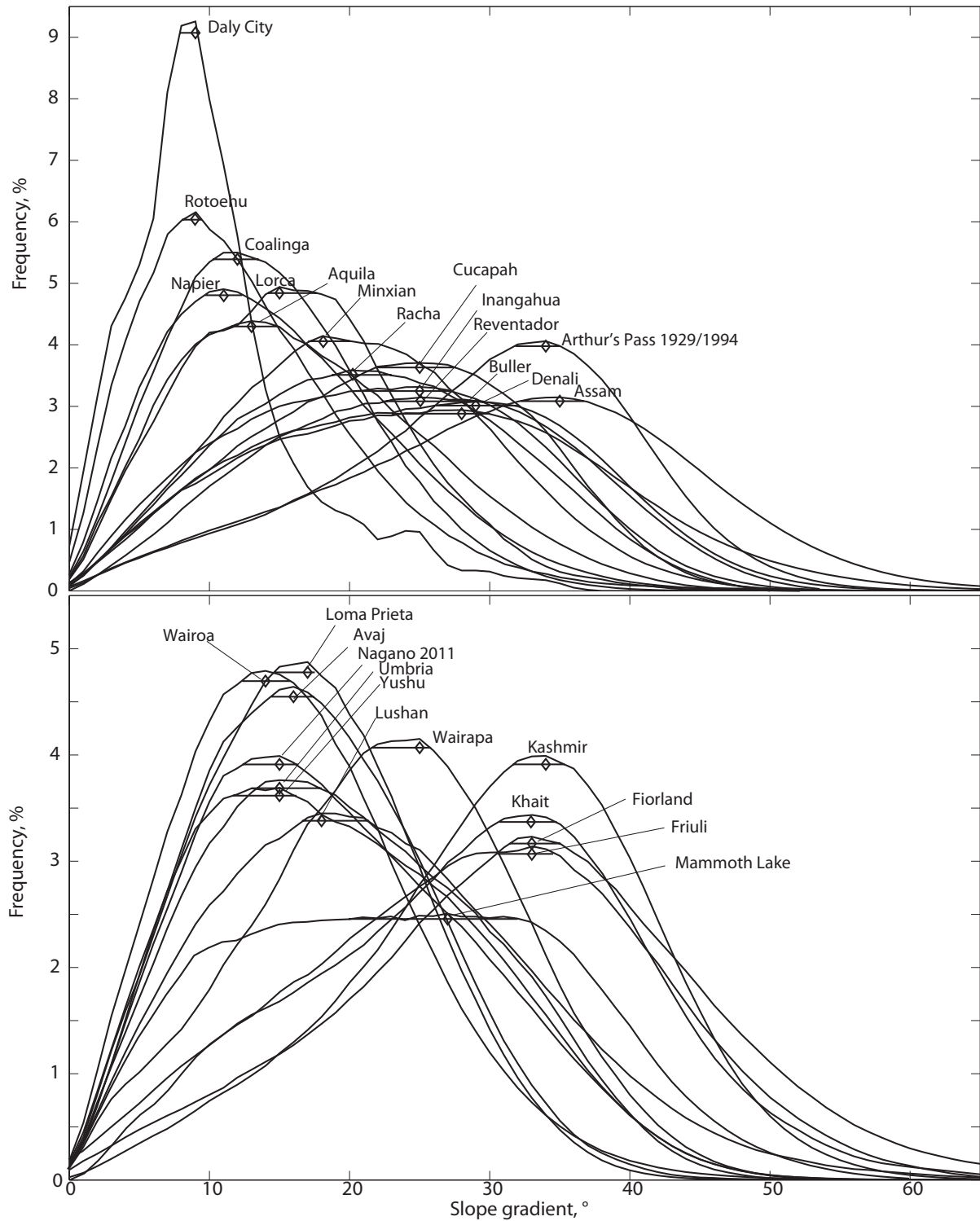


Figure S5. Same as figure 3 of the main text but for the areas of the 29 earthquakes in our compilation for which we do not have comprehensive landslide maps.

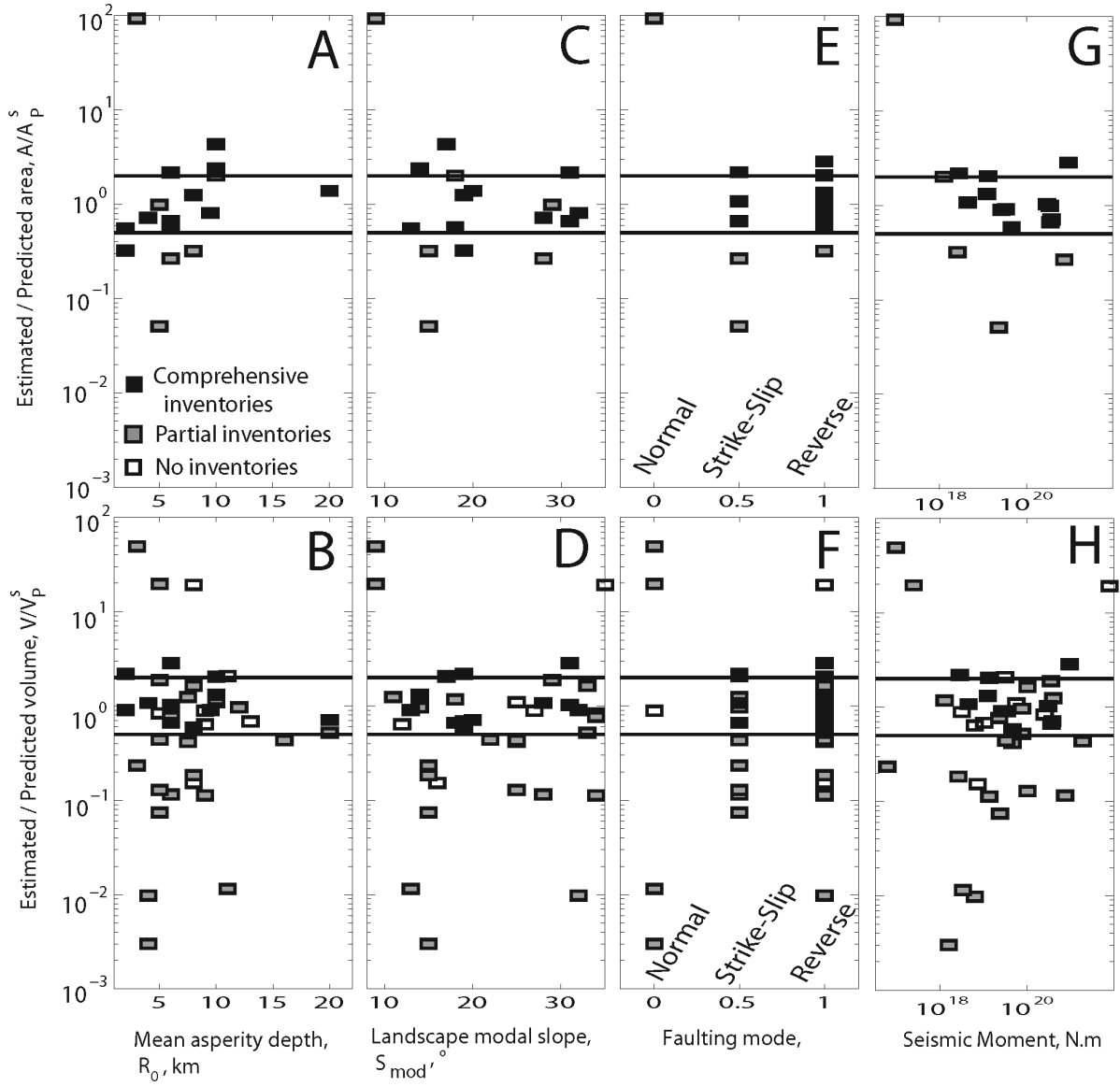


Figure S6. Ratio of estimated total landslide area or volume over predicted total landslide area or volume plotted against asperity mean depth R_0 (A,B), landscape steepness, S_{mod} (C,D), faulting mode (E,F) and seismic moment (G,H). Note the absence of residual correlation in all panels except in panel A, suggesting that the mismatch between data and model prediction arises from factors not included in the model, (such as rock mass strength, hydrological state or shaking duration), and/or from a discrepancy between actual earthquake characteristics and our seismological assumptions in some individual cases (See Section 3). Nevertheless, in the majority of cases, the ratio of estimated and predicted values is close to one, suggesting that for these cases the model assumptions are reasonable and the neglected factors are secondary relative to the ones considered.

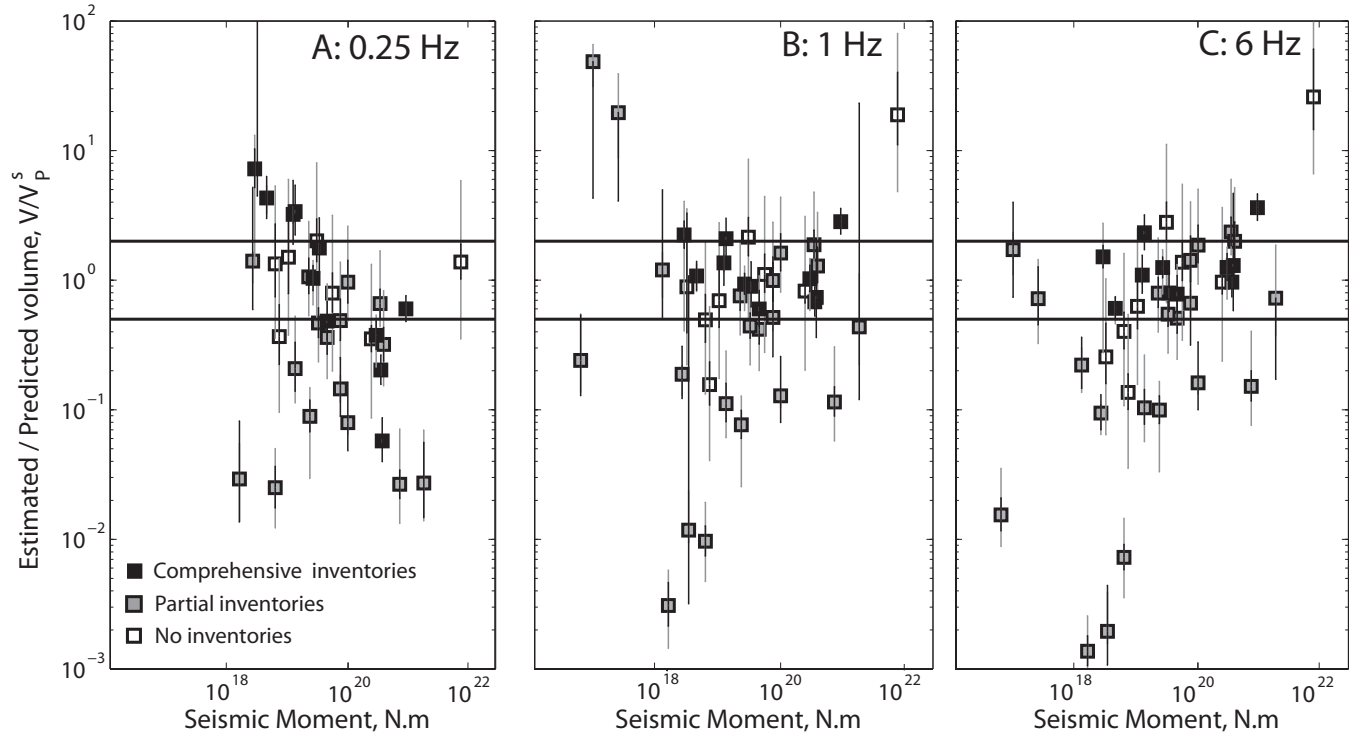


Figure S7. Ratio of estimated total landslide volume over predicted total landslide volume plotted against earthquake seismic moment for different source term frequencies (Eq. 11) A: 0.25Hz, B: 1Hz, C: 6Hz.

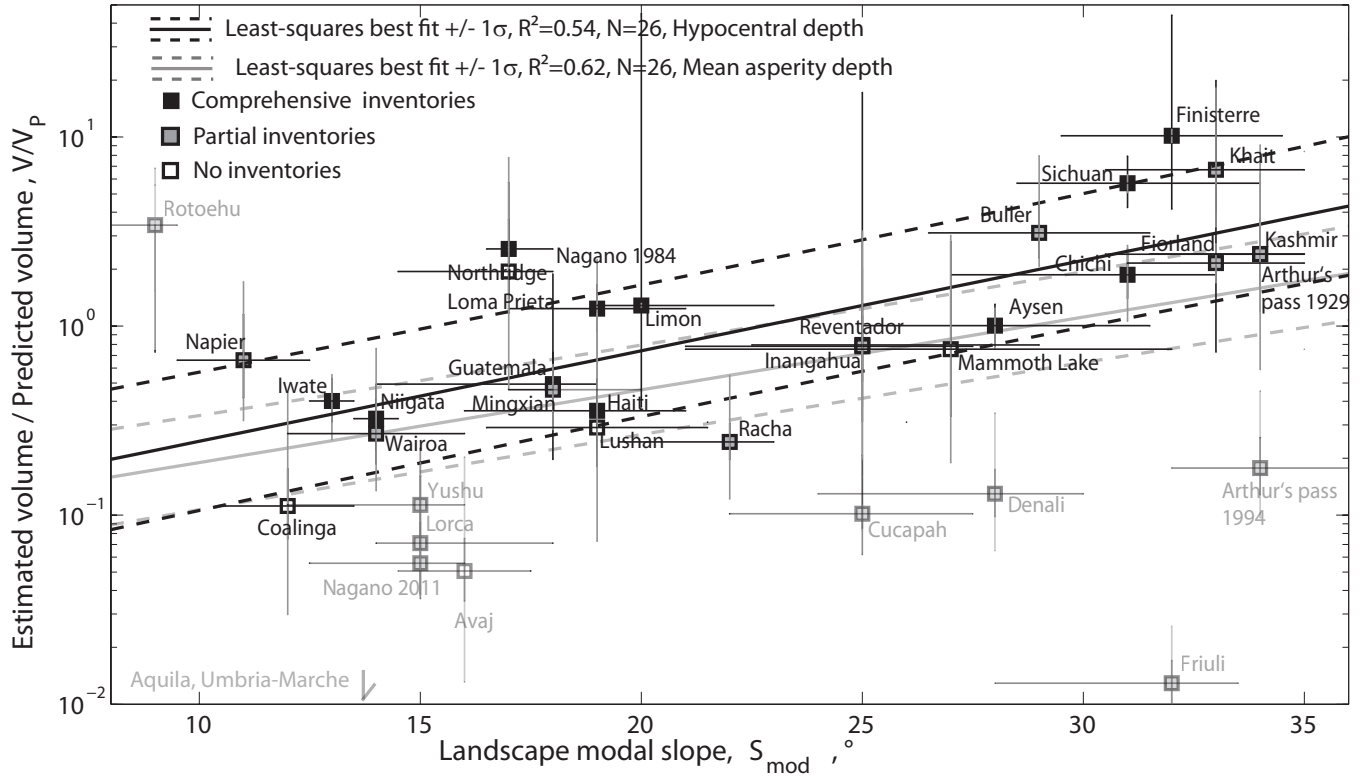


Figure S8. Same as Figure 4 but with source depth set as the hypocentral depth. Therefore some events (e.g. Finisterre, Northridge, Napier, Buller, Khait, etc.) have a strongly reduced prediction and thus a residual larger than in Figure 4 resulting in a stronger control of the landscape modal slope and a larger scatter. For reference the fit obtained in Figure 4 is included.

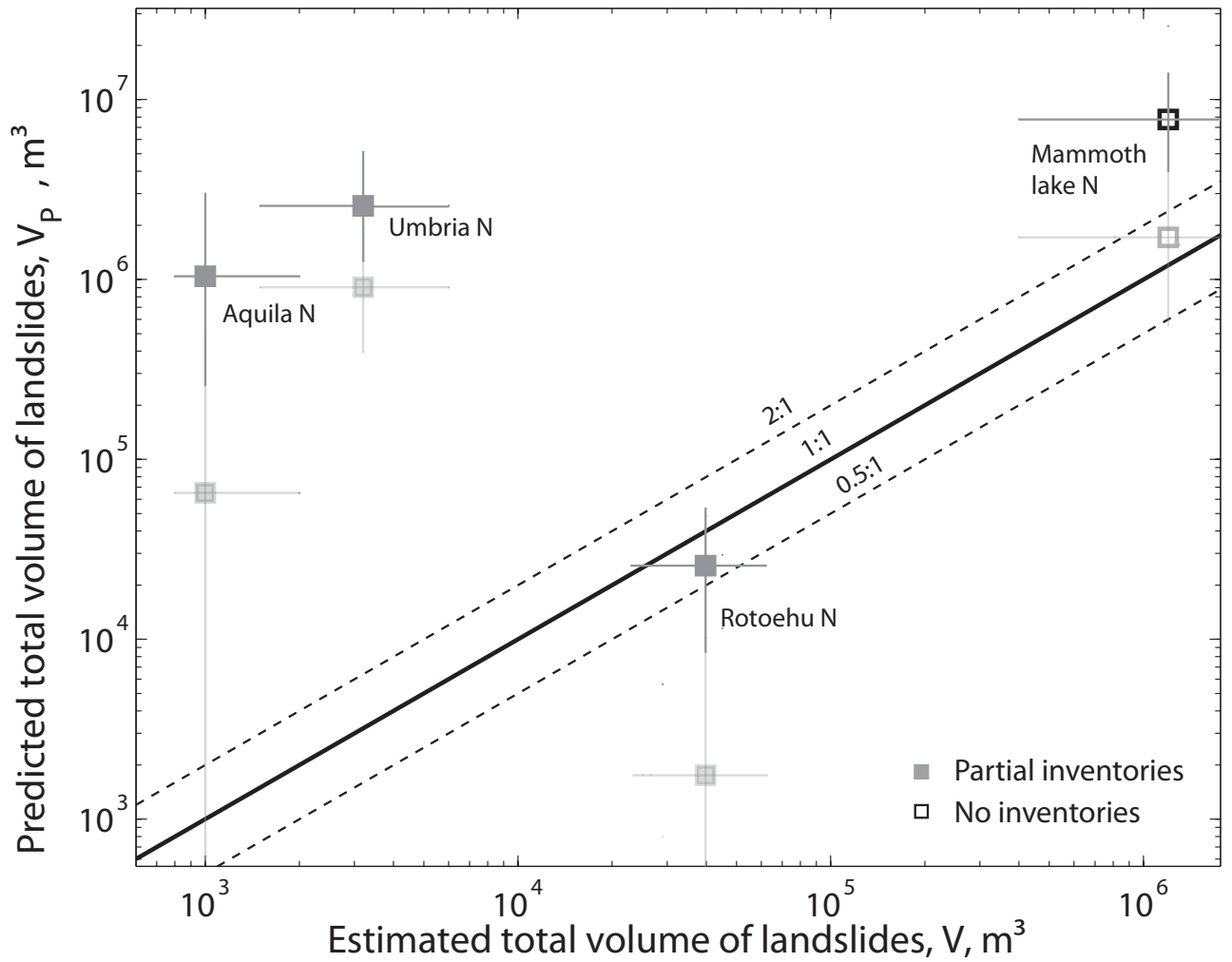


Figure S9. Predicted total landslide volume plotted against estimated volume. In this prediction, normal faults are assumed to produce as much shaking as reverse slip and strike slip mechanisms. For comparison, the model prediction with 30% reduced shaking for normal faults (as in Figure 6) are shown in gray.

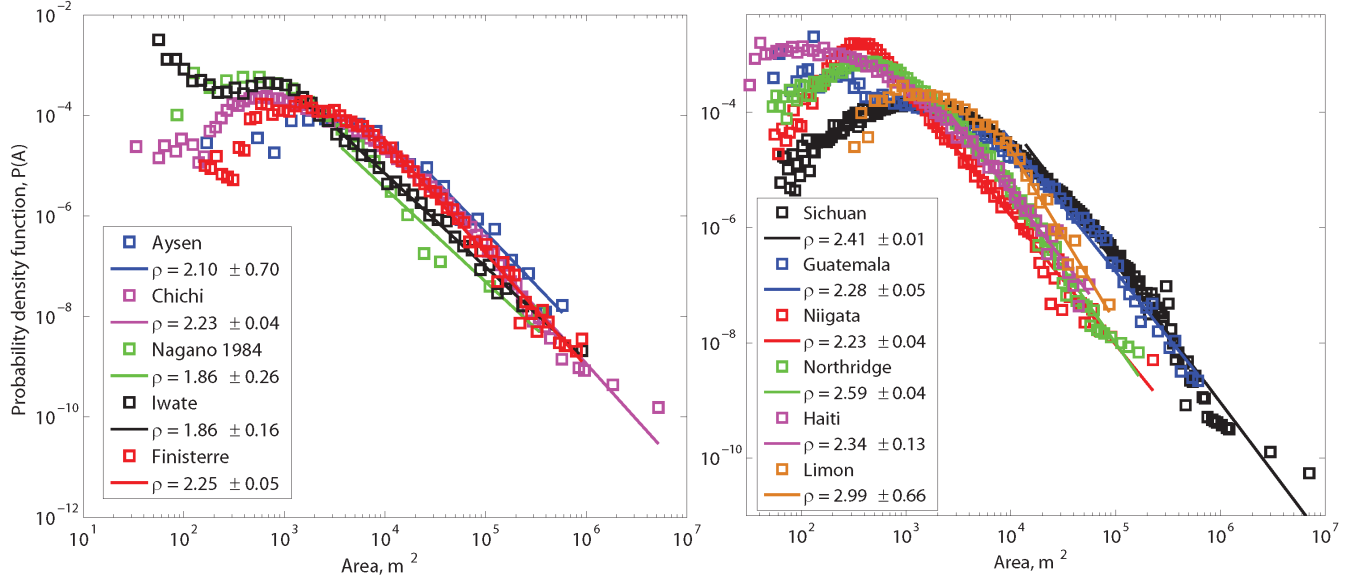


Figure S10. Landslide area-frequency distribution for our 11 exhaustive inventories. Bins were logarithmically spaced between the smallest and largest landslide area of each catalogue, with a number of bins equal to the square root of the number of landslides in the inventory. The decay exponent and its $1 - \sigma$ uncertainty were obtained by least-square minimization of the bins with a probability density 10 times smaller than the probability density at the roll-over to avoid to be biased by the roll-over of the distribution.

Table S2. Summary of the methods used to estimate total landslide volume and its uncertainty from published information. V_A is the total volume and uncertainties estimated from converting into volume available landslide frequency-area distribution. V_P is the total volume estimated as 150% of the sum volume of (N) large, reported landslides, assumed to represent the largest triggered landslides (see Methods). In this case, the uncertainty range is assumed to be defined between 50-270% of V_P . When only area information was given, landslide volumes were derived with area-volume relationships (see Methods). V_M is the total landslide volume based on the volume of the largest landslide and assuming a universal frequency-area distribution (see Methods). V_{max} is the largest landslide volume reported. V_L is the landslide volume estimate reported by other authors (see Suppl. Table 1 for references). For V_M and V_L the uncertainties are deemed large and assumed to extend between 25-400% of the estimated total volume. We consider V_A to be the most accurate, followed by V_P , and then V_M . The value in bold is the one considered further in the study and reported in Table 1. ** indicates that some deep-seated landslides with significant displacement (several meters), but not necessarily with catastrophic failure, were included in the total volume estimate.

Earthquake	V_A (km^3)	V_P (km^3), [N]	V_M (km^3)	V_{max}	$1e6 m^3$	V_L (km^3)
Wairapa 1855	x	0.078 [29]	0.067	11	x	x
Arthur Pass 1929	x	0.15 [6]	0.47	72	0.059	
Buller 1929	x	0.9 [66]	1.47	210	1 - 1.3	
Napier 1930	x	0.25 [9]	0.19	30	x	
Wairoa 1935	x	0.046 [4]	0.12	20	x	
Khait 1949	x	0.48 [15-30?]	0.5	75	x	
Assam 1950	x	x	x	x	47	
Daly City 1957	4.5e-5 [2.9e-5 - 6.1e-5]	x	14e-5	0.03	6.7e-5	
Inangahua 1968	x	0.027 [10]	0.03	5	0.052	
Friuli 1976	x	4.5e-4 [11]	5e-4	0.1	x	
Mammoth Lake 1980	x	x	1e-3	0.2	1.2e-3	
Coalinga 1983	x	x	1e-3	0.2	2e-3	
Reventador 1987	x	x	x	x	0.076 ; 0.12	
Loma Prieta 1989	x	x	0.07	12	0.075	
Racha 1991	x	0.16 [7]**	0.33**	50**	x	
Arthur Pass 1994	x	0.011 [20]	0.011	2	x	
Umbria Marche 1999	3.2e-6 [1.5e-6 - 6e-6]	x	0.8e-6	0.2e-3	x	
Avaj 2002	x	x	5e-4 **	0.1**	x	
Denali 2002	x	0.12 [7]	0.12	20	x	
Fiorland 2003	5.7e-3 [4.5e-3 - 7.2e-3]	x	x	x	x	
Rotoehu 2004	4e-5 (1.8e-5 - 8e-5) !	x	x	x	x	
Kashmir 2005	x	x	0.5	80	x	
Aquila 2009	1e-6 (8e-7 - 2e-6)	x	2e-6	0.5e-3	x	
Cucapah 2010	x	x	2.3e-4	0.05	2.7e-3 [2.4e-3 - 3e-3]	
Yushu 2010	x	x	x	x	3e-3 [1e-3 5e-3]	
Nagano 2011	6e-4 (4.1e-3 8e-4)	x	7.5e-4	0.14	x	
Lorca 2011	3.5e-6 (2e-6 - 8e-6) !	x	x	x	x	
Lushan 2013	x	x	6e-3	1.2	x	
Mingxian 2013	8e-4 [5e-4 1.1e-3]	x	13e-4	0.25	x	